Project 1: The Hybrid Deep Learning Model for Identification of Attention-Deficit/Hyperactivity D

	Content
Project Title	The Hybrid Deep Learning Model for Identification of Attention-Deficit/Hyperactivity Disorder Using EEG
Category	Neuro
Problem Statement	Attention Deficit Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder characterized by inattention, hyperactivity, and impulsive behavior. Current diagnosis relies largely on behavioral assessments, clinical observations, and questionnaires, which can be subjective and time-consuming. This project aims to develop an AI-based system that analyzes EEG brain signals to automatically identify ADHD-related brain activity patterns and assist clinicians in early and objective diagnosis.
Tech Stack	Python, TensorFlow/PyTorch, NumPy, Pandas, MNE-Python, CNN, LSTM
Max Groups	1

Project 2: Classification of attention deficit/hyperactivity disorder based on EEG signals using a

	Content
Project Title	Classification of attention deficit/hyperactivity disorder based on EEG signals using a EEG-Transformer model
Category	Neuro
Problem Statement	Develop a Transformer-based deep learning system for automatic ADHD detection using EEG brain signals. The project addresses the challenge of subjective and time-consuming ADHD diagnosis by learning attention-related brain activity patterns directly from EEG recordings. Students will preprocess EEG data, train an EEG-Transformer model, and classify ADHD patients and healthy controls. The system aims to support clinicians through objective and AI-assisted screening.
Tech Stack	Python, TensorFlow/PyTorch, MNE-Python, NumPy, Pandas, Scikit-Learn, Transformer Networks, Matplotlib
Max Groups	1

Project 3: A Novel Approach to Schizophrenia Detection: Optimized Preprocessing and Deep L

	Content
Project Title	A Novel Approach to Schizophrenia Detection: Optimized Preprocessing and Deep Learning Analysis of Multichannel EEG Data
Category	Neuro
Problem Statement	The project addresses the challenge of subjective and time-consuming schizophrenia diagnosis, which currently relies on clinical interviews and behavioral assessments. By analyzing EEG recordings, the system aims to identify abnormal brain activity patterns associated with schizophrenia and provide an objective AI-assisted screening tool for early detection and diagnosis support.Develop a deep learning-based system for automatic schizophrenia detection using multichannel EEG brain signals.
Tech Stack	Python, MNE-Python, NumPy, Pandas, TensorFlow/PyTorch, CNN, Scikit-Learn, Matplotlib
Max Groups	1

Project 4: CNN feature and classifier fusion on novel transformed image dataset for dysgraphia

	Content
Project Title	CNN feature and classifier fusion on novel transformed image dataset for dysgraphia diagnosis in children
Category	Neuro
Problem Statement	Develop an automated dysgraphia diagnosis system using handwriting images and deep learning. The project addresses the challenge of subjective and time-consuming dysgraphia diagnosis by applying transfer learning, CNN feature extraction, ensemble learning, and feature fusion techniques on handwriting tasks such as words, pseudo-words, difficult words, and sentences. The objective is to automatically distinguish dysgraphic handwriting from normal handwriting with high accuracy.
Tech Stack	Python, TensorFlow, Keras, DenseNet201, SVM, Random Forest, AdaBoost, OpenCV, Transfer Learning
Max Groups	1

Project 5: Neurological state changes indicative of ADHD in children learned via EEG-based LS

	Content
Project Title	Neurological state changes indicative of ADHD in children learned via EEG-based LSTM networks
Category	Neuro
Problem Statement	The project addresses the challenge of subjective ADHD diagnosis by identifying neurological state changes associated with attention deficits, impulsivity, and hyperactivity. Unlike traditional methods that rely on questionnaires and behavioral observations, the system analyzes temporal patterns in EEG recordings to learn how brain activity changes over time and automatically classify ADHD and healthy subjects. The objective is to provide an objective AI-assisted screening tool for early ADHD detection. Develop an LSTM-based deep learning system for ADHD detection using EEG brain signals.
Tech Stack	Python, MNE-Python, NumPy, Pandas, TensorFlow/PyTorch, LSTM, Scikit-Learn, Matplotlib
Max Groups	1

Project 6: CVSM: Contrastive Vocal Similarity Modeling

	Content
Project Title	CVSM: Contrastive Vocal Similarity Modeling
Category	Acoustic
Problem Statement	Contrastive self-supervised learning for music signal representation focused on vocal/musical similarity. Uses label-informed (artist identity) and label-agnostic (artificial mixture) contrastive protocols.
Tech Stack	Contrastive learning (SimCLR/MoCo), audio/music signal processing, representation learning, linear probing, PyTorch
Max Groups	1

Project 7: MT2KD: Toward a General-Purpose Encoder for Speech, Speaker, and Audio Event

	Content
Project Title	MT2KD: Toward a General-Purpose Encoder for Speech, Speaker, and Audio Events
Category	Acoustic
Problem Statement	Two-stage multi-task framework: multi-teacher Knowledge Distillation (Whisper+PANN+ECAPA-TDNN !' Zipformer student) then supervised fine-tuning for ASR, Audio Tagg jointly. 66M parameters.
Tech Stack	ASR, knowledge distillation, multi-task learning, Conformer/Zipformer, speaker verification basics, PyTorch or Icefall
Max Groups	1

Project 8: A Generative Model for Cetacean Whistles Using Variational Autoencoder and Mixtu

	Content
Project Title	A Generative Model for Cetacean Whistles Using Variational Autoencoder and Mixture of Gaussians
Category	Acoustic
Problem Statement	VAE-GMM framework to model false killer whale whistle signals for biomimetic underwater communication. Up to 10,000 Gaussian components via EM algorithm. Enables unsupervised clustering and conditional whistle generation.
Tech Stack	VAE, Gaussian Mixture Models, EM algorithm, spectrogram image processing, signal processing, Python/PyTorch or MATLAB
Max Groups	1

Project 9: USEF-TSE: Universal Speaker Embedding Free Target Speaker Extraction

	Content
Project Title	USEF-TSE: Universal Speaker Embedding Free Target Speaker Extraction
Category	Acoustic
Problem Statement	Universal framework for target speaker extraction without speaker embeddings. Uses CMHA to extract frame-level speaker features from reference speech. SOTA SI-SDRi: 23.3dB (WSJ0-2mix), 17.6dB (WHAM!), 16.1dB (WHAMR!).
Tech Stack	Speech separation (TasNet, SepFormer, TF-GridNet), attention mechanisms, time-domain and TF-domain audio processing, PyTorch or SpeechBrain
Max Groups	1

Project 10: A novel multi-scale network intrusion detection model with transformer

	Content
Project Title	A novel multi-scale network intrusion detection model with transformer
Category	Cyber
Problem Statement	This project implements a deep learning-based Intrusion Detection System that detects normal and malicious network traffic using multi-scale traffic feature extraction and Transformer-based classification. The system preprocesses network traffic data, converts it into matrix form, extracts multi-scale features using convolution branches, applies Transformer encoders for feature learning, and classifies traffic into normal or attack categories.
Tech Stack	Python, PyTorch, NumPy, Scikit-learn, Matplotlib, Pandas, One-hot Encoding, Min-Max Normalization, CNN, Dilated Convolution, Transformer Encoder, Multi-Head Self-Attention, Focal Loss.
Max Groups	1

Project 11: Improving phishing email detection performance through deep learning with adaptiv

	Content
Project Title	Improving phishing email detection performance through deep learning with adaptive optimization
Category	Cyber
Problem Statement	This project implements a deep learning-based phishing email detection system that classifies emails as Phishing or Safe/Legitimate. The system uses BERT embeddings to understand email text, CNN to extract local text patterns, GRU to capture sequential dependencies, multi-head attention to focus on important phishing-related words or phrases, and an optimization method to tune model hyperparameters for improved detection performance.
Tech Stack	Python, Google Colab Pro, TensorFlow/Keras, NumPy, Pandas, BERT, BERT Tokenizer, CNN-1D, GRU, Multi-Head Attention, Dropout, L2 Regularization, Mountain Gazelle Optimizer-style hyperparameter tuning
Max Groups	1

Project 12: Efficient malware detection using NLP and deep learning model

	Content
Project Title	Efficient malware detection using NLP and deep learning model
Category	Cyber
Problem Statement	This project implements a malware detection system that classifies files as Benign or Malicious using behavioural analysis. The system collects kernel/API function calls from sandbox analysis, converts the function-call sequence into BERT-based vector embeddings, and passes the resulting feature vector into a multi-layer perceptron model for binary classification.
Tech Stack	Python, Google Colab, PyTorch, BERT tokenizer/model, Pandas, NumPy, Scikit-learn, Cuckoo Sandbox, Virtual Machine/Linux environment, kernel API/function-call analysis, vector embeddings
Max Groups	1

Project 13: XRan: Explainable deep learning-based ransomware detection using dynamic analy

	Content
Project Title	XRan: Explainable deep learning-based ransomware detection using dynamic analysis
Category	Cyber
Problem Statement	This project implements an explainable ransomware detection system that classifies executable behavior as Ransomware, Benign, or optionally General Malware. The system uses dynamic analysis reports to extract API call sequences, DLL sequences, and mutex/mutual exclusion sequences. These sequences are merged and given to a 2-layer CNN model for ransomware detection. LIME and SHAP are used to explain which API calls, DLLs, or mutexes influenced the model's decision.
Tech Stack	Python, PyTorch, Pandas, NumPy, Scikit-learn, BERT, BERT Tokenizer, CNN, SMOTE, ReLU, MaxPooling
Max Groups	1

Project 14: A structure-based framework for selective inhibitor design and optimization

	Content
Project Title	A structure-based framework for selective inhibitor design and optimization
Category	Drug Discovery
Problem Statement	This project uses AI, protein structural analysis, molecular modeling, and structure-based drug design to discover selective inhibitors. Its main goal is to develop molecules that strongly bind to the desired target protein while avoiding similar proteins, thereby reducing side effects. The framework combines binding-pocket analysis, affinity and selectivity prediction, and lead optimization. It also performs multi-objective optimization for binding strength, drug-likeness, synthetic accessibility, and safety.
Tech Stack	"TensorFlow/Keras or PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, PyTorch, Graph Neural Networks (GNNs) Graph Attention Networks (GATs) Transformers Diffusion Models
Max Groups	1

Project 15: Explainable drug sensitivity prediction through cancer pathway enrichment

	Content
Project Title	Explainable drug sensitivity prediction through cancer pathway enrichment
Category	Drug Discovery
Problem Statement	an explainable deep learning framework that predicts how sensitive a cancer cell is to a specific anticancer drug. PathDSP transforms genomic and drug information into biologically meaningful pathway features before prediction. This allows researchers to understand why a drug is predicted to work for a particular cancer rather than only obtaining a prediction. ""The challenge is to accurately predict: Will a particular drug be effective against a specific cancer? using genomic information while also providing biological explanations for the prediction.
Tech Stack	"Python, Scikit-learn,, Pandas, NumPy, Matplotlib, Plotly, GSEApy (Gene Set Enrichment) RDKit Biopython, Deep Neural Networks, GNN"
Max Groups	1

Project 16: Fast and Flexible Protein Design Using Deep Graph Neural Networks

	Content
Project Title	Fast and Flexible Protein Design Using Deep Graph Neural Networks
Category	Drug Discovery
Problem Statement	"The project develops an AI-assisted inverse protein-design system that generates amino-acid sequences compatible with a user-specified protein topology. The target protein structure is converted into a graph in which residues are represented as nodes and spatial interactions are represented as edges. A pretrained ProteinSolver graph neural network predicts suitable amino acids for unknown or redesignable residue positions. Multiple candidate sequences are generated, ranked using sequence and model-confidence measures, and computationally validated to determine whether they are likely to reproduce the required three-dimensional structure.The model was trained using more than 70 million protein sequences associated with over 80,000 experimentally determined protein structures."
Tech Stack	"Python PyTorch Geometric NumPy Pandas Matplotlib Protein Data Bank (PDB) UniParc Gene3D , GNN"
Max Groups	1

Project 17: Perceiver CPI: a nested cross-attention network for compound–protein interaction p

	Content
Project Title	Perceiver CPI: a nested cross-attention network for compound–protein interaction prediction
Category	Drug Discovery
Problem Statement	"The Attention-Based Compound–Protein Interaction (CPI) Prediction for Drug Discovery project focuses on developing a deep learning model that predicts whether a chemical compound (drug molecule) can effectively interact with a target protein. Identifying these interactions is one of the most important steps in drug discovery because a drug must bind to a disease-related protein to produce a therapeutic effect. The proposed attention-based model automatically learns important features from:molecular structures of compounds, and amino acid sequences of proteins,to predict whether they interact, thereby reducing both cost and development time. ""The attention layer is the core component of the project. Instead of treating every atom or amino acid equally, the attention mechanism learns which parts of the compound and protein are most important for binding.""
Tech Stack	"TensorFlow/Keras or PyTorch, RDKit, Scikit-learn, Pandas NumPy, Matplotlib, Plotly, Streamlit "
Max Groups	1

Project 18: compound–protein interaction prediction

	Content
Project Title	compound–protein interaction prediction
Category	Nutra
Problem Statement	People respond differently to food because of variations in metabolism, blood biomarkers, BMI, age, gender, and lifestyle. However, most existing nutrition applications provide general diet plans based mainly on calorie intake, height, weight, and BMI, without properly considering an individual's biochemical condition. The base study establishes a relationship between standard blood biochemistry, BMI, and dietary patterns using deep neural networks, with important features such as ALT, AST, uric acid, glucose, cholesterol, energy intake, fat, sugar, protein, carbohydrates, caffeine, and vitamins. However, the study mainly focuses on weight-category classification and dietary-pattern analysis rather than generating interactive personalized recommendations. Therefore, BioNutriGPT is proposed as an AI-based system that analyses blood biomarkers, BMI, food intake, demographic information, and lifestyle factors to predict nutritional and weight-related risks. The system will identify the factors influencing each prediction and generate understandable, personalized nutrition recommendations in natural language for users and nutrition professionals.
Tech Stack	Python, NumPy, Pandas,Scikit-learn, Deep Backpropagation Neural Network
Max Groups	1

Project 19: Large language models in food science: Innovations, applications, and future

	Content
Project Title	Large language models in food science: Innovations, applications, and future
Category	Nutra
Problem Statement	Existing nutrition and food recommendation systems mainly provide calorie tracking and basic dietary advice but lack intelligent understanding of food-related knowledge. Current solutions struggle to analyze nutritional content, food safety risks, ingredient substitutions, and personalized dietary requirements simultaneously. There is a need for an AI-powered system that can understand natural language queries and provide context-aware food insights. The proposed FoodInsightAI leverages Large Language Models (LLMs) to analyze food composition, nutrition, recipes, allergens, and food safety information. The system generates personalized dietary recommendations based on user preferences, health conditions, and nutritional requirements. It also assists users with recipe generation, ingredient substitutions, and food quality awareness. This approach aims to improve decision-making, promote healthier eating habits, and enhance food safety through intelligent conversational AI.
Tech Stack	Python, NumPy, Pandas,PyTorch, Docker, Hugging Face Spaces,Retrieval-Augmented Generation (RAG), FAISS/ChromaDB (optional), ransformers, Tokenizers
Max Groups	1

Project 20: "Integrated Molecular Docking and Network-Based Analysis Reveals Multitarget Int

	Content
Project Title	"Integrated Molecular Docking and Network-Based Analysis Reveals Multitarget Interaction Patterns of Nutraceutical Compounds in Intervertebral Disc Degeneration"
Category	Nutra
Problem Statement	Intervertebral Disc Degeneration (IVDD) is a major cause of chronic low back pain caused by inflammation and extracellular matrix (ECM) degradation. Current treatment approaches are limited in addressing the disease's complex molecular mechanisms. Nutraceutical compounds such as curcumin, quercetin, resveratrol, melatonin, and baicalein have shown potential therapeutic effects. However, their multitarget interactions with IVDD-associated proteins remain unclear. This study integrates molecular docking with network-based analysis to identify key protein interactions and biological pathways. The findings provide a computational framework for discovering potential therapeutic targets for IVDD.
Tech Stack	"Python, ChemDraw Ultra ChemBio3D Ultra Open Babel Cytoscape cytoHubba Plugin STRING Database Gene Ontology (GO) Analysis KEGG Pathway Analysis
Max Groups	1

Project 21: "Population-Specific Glucose Prediction in Diabetes Care With Transformer-Based

	Content
Project Title	"Population-Specific Glucose Prediction in Diabetes Care With Transformer-Based Deep Learning on the Edge"
Category	Nutra
Problem Statement	Diabetes management requires continuous monitoring of blood glucose levels to prevent dangerous events such as hypoglycemia and hyperglycemia. Existing prediction systems are often designed for individual patients, have limited generalization across populations, and rely heavily on cloud computing, leading to latency and privacy concerns. This project aims to develop a Transformer-based population-specific blood glucose prediction system that leverages Continuous Glucose Monitoring (CGM) data and patient demographic information for accurate multi-horizon glucose forecasting. The proposed model utilizes a Temporal Fusion Transformer (TFT) to capture temporal dependencies and personalize predictions based on static patient characteristics. The trained model is deployed on a low-power wearable edge device to provide real-time glucose predictions without requiring continuous internet connectivity. This enables timely medical interventions, reduces the risk of adverse glycemic events, and improves diabetes management for both Type 1 and Type 2 diabetes patients.
Tech Stack	"Python PyTorch Temporal Fusion Transformer (TFT) CMSIS-DSP Library"
Max Groups	1

Project 22: "OralAbsPredict: A data-driven framework to predict human intestinal absorption (H

	Content
Project Title	"OralAbsPredict: A data-driven framework to predict human intestinal absorption (HIA) and human oral bioavailability (HOB) from chemical structures"
Category	Nutra
Problem Statement	Poor oral absorption is one of the primary reasons for drug failure during the early stages of drug discovery, leading to increased development time and cost. Experimental evaluation of Human Intestinal Absorption (HIA) and Human Oral Bioavailability (HOB) is expensive, time-consuming, and resource-intensive. Existing computational methods often suffer from limited datasets, class imbalance, and poor interpretability. This project aims to develop an AI-driven framework that predicts HIA and HOB directly from molecular structures represented as SMILES strings. The system employs machine learning and molecular descriptors to accurately classify orally active compounds before synthesis. It also provides feature interpretation using explainable AI techniques and identifies important molecular substructures responsible for oral absorption. This approach enables rapid screening of drug candidates, reducing experimental costs and accelerating drug discovery.
Tech Stack	"Python Scikit-learn RDKit Mordred Descriptor Calculator CatBoost LightGBM Support Vector Machine (SVM) Random Forest (RF) Balanced Random Forest (BRF) Logistic Regression SHAP (Explainable AI)"
Max Groups	1